# 项目 5：大中型园区网络安全架构综合项目

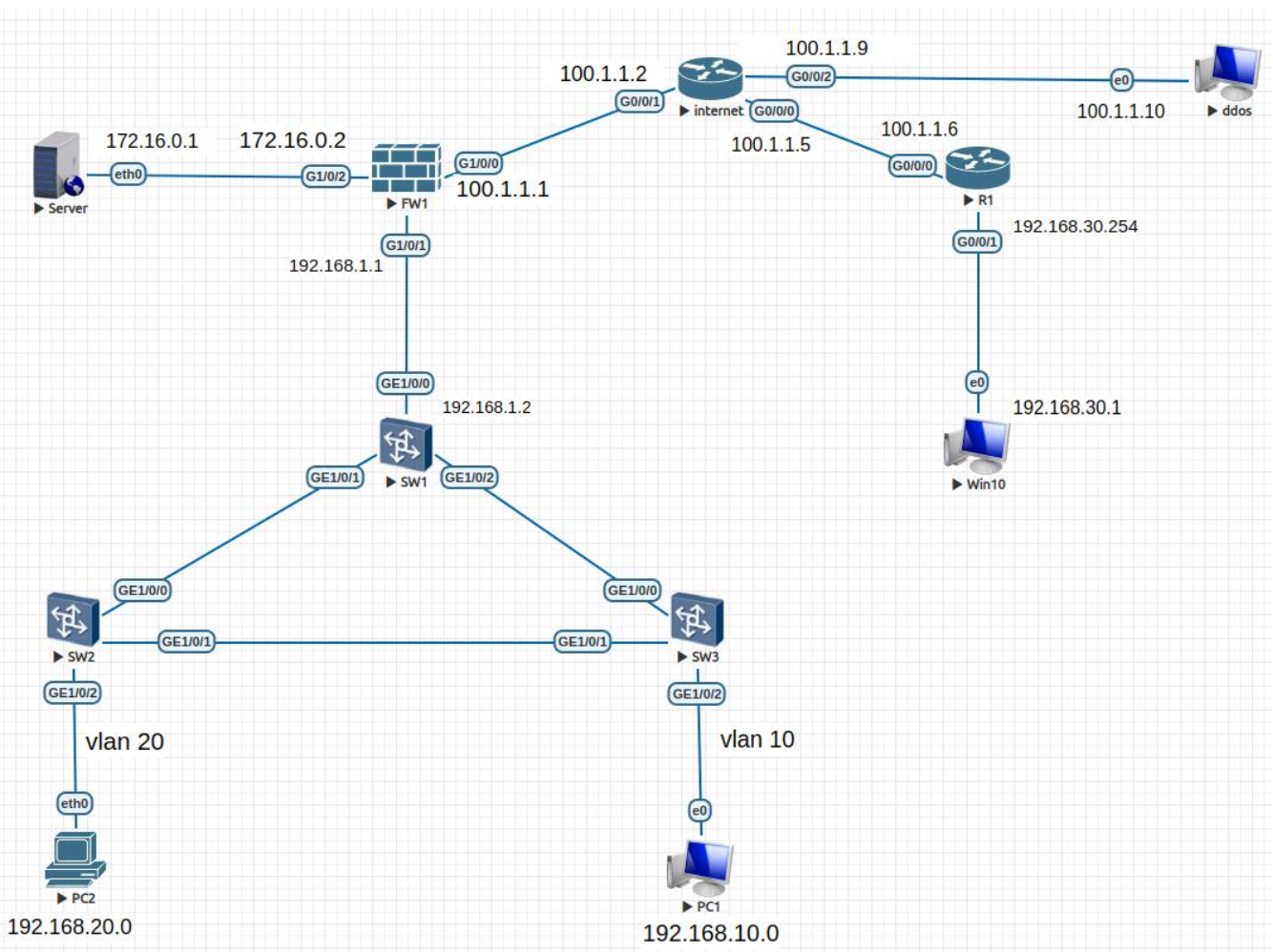
## 【任务描述】

某公司随着业务规模扩大，网络架构日益复杂，当前面临多重网络问题与需求：一方面，公司总部与分支之间缺乏安全可靠的内网互访通道，数据传输存在泄露风险；另一方面，内部服务器需向外部提供服务，但缺乏有效的地址映射机制，外网用户无法正常访问；更紧急的是，近期公司网络频繁遭遇 DDoS 攻击，服务器运行稳定性受严重影响，部分业务出现中断。此外，公司内网终端 IP 地址仍依赖手动配置，不仅效率低下，还易因地址冲突引发网络故障，同时内网用户访问互联网时，也存在地址转换不畅的问题。

为解决上述问题，保障公司网络安全、稳定、高效运行，现安排工程师小李完成以下任务：

1. 依据给定的网络拓扑与防火墙规划表，完成网络设备（路由器、交换机、防火墙等）的基础配置，包括设备命名、IP 地址配置、VLAN 划分、接口类型设置及路由协议部署，确保网络基础架构通畅；
2. 配置 DHCP 服务与监听功能，实现内网终端自动获取 IP 地址，避免地址冲突，提升终端入网效率；
3. 配置源 NAT 策略，解决内网用户访问互联网的地址转换问题，保障外网访问需求；
4. 部署 VPN（GRE 隧道 + IPSec 加密），建立总部与分支之间的安全通信通道，确保跨区域内网数据传输的保密性与完整性；
5. 配置 NAT Server，将内部服务器地址映射到公网地址特定端口，实现外网用户对内部服务器服务的正常访问；
6. 配置 DDoS 攻击防范策略，包括流量统计、基线学习、攻击检测与防御功能，抵御 DDoS 攻击，保障服务器与网络的稳定运行。

拓扑图：




## 【IP 规划】

| 设备名称 | 端口 | 接口类型 | 接口所在网段 | 网关 |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| INTERNET | G0/0/1 | | 100.1.1.2/30 | |
| | G0/0/0 | | 100.1.1.5/30 | |
| FW1 | G1/0/0 | | 100.1.1.1/30 | |
| | G1/0/1 | | 192.168.1.1/30 | |
| | G1/0/2 | | 172.16.0.2/24 | |
| SW1 | GE1/0/0 | Access | Vlan1000:192.168.1.2/30 | |
| | | | Vlan10:192.168.10.254/24 | |
| | | | Vlan20:192.168.20.254/24 | |
| | GE1/0/1 | Trunk | Vlan 10,20 | |
| | GE1/0/2 | Trunk | Vlan 10,20 | |
| R1 | G0/0/0 | | 100.1.1.6/30 | |
| | G0/0/1 | | 192.168.30.254/24 | |
| | G0/0/2 | | 10.1.1.9/30 | |
| Win10 | E0 | | 192.168.30.1/24 | 192.168.30.254 |
| PC1 | Eth0 | | 192.168.10.0/24 | 192.168.10.254 |
| PC2 | Eth0 | | 192.168.20.0/24 | 192.168.20.254 |
| Server1 | Eth0 | | 172.16.0.1/24 | 172.16.0.2 |
| ddos | E0 | | 100.1.1.10/30 | 100.1.1.9 |

# 【任务实施】

## 一. 基础配置

1. 根据拓扑和规划表修改设备名称，配置 IP 地址，划分 vlan。

```
[R1]display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(E): E-Trunk down
The number of interface that is UP in Physical is 8
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 3
The number of interface that is DOWN in Protocol is 5

Interface                         IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0              100.1.1.6/30         up         up
GigabitEthernet0/0/1              192.168.30.254/24    up         up
GigabitEthernet0/0/2              unassigned           up         down
```

```
[internet]display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(E): E-Trunk down
The number of interface that is UP in Physical is 8
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 3
The number of interface that is DOWN in Protocol is 5

Interface                         IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0              100.1.1.5/30         up         up
GigabitEthernet0/0/1              100.1.1.2/30         up         up
GigabitEthernet0/0/2              unassigned           up         down
GigabitEthernet0/0/3              unassigned           up         down
```

```
[SW1]display  ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 3
The number of interface that is UP in Protocol is 1
The number of interface that is DOWN in Protocol is 4
Interface                   IP Address/Mask    Physical Protocol VPN
MEth0/0/0                   unassigned         up       down     --
NULL0                       unassigned         up       up(s)    --
Vlanif10                    192.168.10.254/24  down     down     --
Vlanif20                    192.168.20.254/24  down     down     --
Vlanif1000                  192.168.1.2/30     down     down     --
[SW1]
```

```
[FW1]display ip interface  brief
2025-09-11 06:30:35.610
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 8
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 5
The number of interface that is DOWN in Protocol is 3

Interface                         IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0(10G)         192.168.0.1/24       up         up
GigabitEthernet1/0/0(10G)         100.1.1.1/30         up         up
GigabitEthernet1/0/1(10G)         192.168.1.1/30       up         up
GigabitEthernet1/0/2(10G)         unassigned           up         down
GigabitEthernet1/0/3(10G)         unassigned           up         down
GigabitEthernet1/0/4(10G)         unassigned           up         down
NULL0                             unassigned           up         up(s)
Virtual-if0                       unassigned           up         up(s)
```

2. 分公司出口网关配置去往外网的默认路由。

```
[R1]
[R1]ip route-static 0.0.0.0 0.0.0.0 100.1.1.5
[R1]
```

3. 总部防火墙配置去往外网的默认路由。

```
[FW1]
[FW1]ip route-static 0.0.0.0 0.0.0.0 100.1.1.2
[FW1]
```

4. 交换机配置接口类型。

```
[SW1]interface  GE 1/0/0
[SW1-GE1/0/0]
[SW1-GE1/0/0]port default vlan 1000
[SW1-GE1/0/0]
[SW1-GE1/0/0]quit
[SW1]
[SW1]interface  GE 1/0/1
[SW1-GE1/0/1]
[SW1-GE1/0/1]port link-type trunk
[SW1-GE1/0/1]
[SW1-GE1/0/1]port trunk allow-pass vlan 10 20
[SW1-GE1/0/1]
[SW1-GE1/0/1]quit
[SW1]
[SW1]interface  GE 1/0/2
[SW1-GE1/0/2]
[SW1-GE1/0/2]port link-type trunk
[SW1-GE1/0/2]
[SW1-GE1/0/2]port trunk allow-pass vlan 10 20
[SW1-GE1/0/2]
```

```
[SW2]VLAN  batch 10 20
[SW2]
[SW2]int g1/0/0
[SW2-GE1/0/0]
[SW2-GE1/0/0]port link-type trunk
[SW2-GE1/0/0]
[SW2-GE1/0/0]port trunk allow-pass vlan 10 20
[SW2-GE1/0/0]
[SW2-GE1/0/0]quit
[SW2]
[SW2]int g1/0/1
[SW2-GE1/0/1]
[SW2-GE1/0/1]port link-type trunk
[SW2-GE1/0/1]
[SW2-GE1/0/1]port trunk allow-pass vlan 10 20
[SW2-GE1/0/1]
[SW2-GE1/0/1]quit
[SW2]
[SW2]int g1/0/2
[SW2-GE1/0/2]
[SW2-GE1/0/2]port default vlan 20
```

```
[SW3]vlan batch 10 20
[SW3]
[SW3]int g1/0/0
[SW3-GE1/0/0]port link-type trunk
[SW3-GE1/0/0]
[SW3-GE1/0/0]port trunk allow-pass vlan 10 20
[SW3-GE1/0/0]
[SW3-GE1/0/0]int g1/0/1
[SW3-GE1/0/1]
[SW3-GE1/0/1]port link-type trunk
[SW3-GE1/0/1]
[SW3-GE1/0/1]port trunk allow-pass vlan 10 20
[SW3-GE1/0/1]
[SW3-GE1/0/1]int g1/0/2
[SW3-GE1/0/2]
[SW3-GE1/0/2]port default vlan 10
[SW3-GE1/0/2]
[SW3-GE1/0/2]int g1/0/4
[SW3-GE1/0/4]
[SW3-GE1/0/4]port default vlan 10
[SW3-GE1/0/4]
```

5. 开启生成树协议，防止环路。（以 SW3 为例）

```
[SW3]
[SW3]stp enable
[SW3]
[SW3]stp mode  rstp
[SW3]
```

6. 总部内网配置 ospf 路由协议实现全网互通。

```
[SW1]
[SW1]ospf 1
[SW1-ospf-1]
[SW1-ospf-1]area 0
[SW1-ospf-1-area-0.0.0.0]network  192.168.10.0 0.0.0.255
[SW1-ospf-1-area-0.0.0.0]
[SW1-ospf-1-area-0.0.0.0]network  192.168.20.0 0.0.0.255
[SW1-ospf-1-area-0.0.0.0]
[SW1-ospf-1-area-0.0.0.0]network 192.168.1.2 0.0.0.3
[SW1-ospf-1-area-0.0.0.0]
```

```
[FW1]
[FW1]ospf 1
[FW1-ospf-1]default-route-advertise always  # 发布默认路由
[FW1-ospf-1]
[FW1-ospf-1]area 0
[FW1-ospf-1-area-0.0.0.0]
[FW1-ospf-1-area-0.0.0.0]network 192.168.1.1 0.0.0.3
[FW1-ospf-1-area-0.0.0.0]
[FW1-ospf-1-area-0.0.0.0]
```

7. 配置接口服务管理和划分安全区域。

```
[FW1]
[FW1]interface  GigabitEthernet 1/0/0
[FW1-GigabitEthernet1/0/0]
[FW1-GigabitEthernet1/0/0]service-manage enable
[FW1-GigabitEthernet1/0/0]
[FW1-GigabitEthernet1/0/0]service-manage ping permit
[FW1-GigabitEthernet1/0/0]
[FW1-GigabitEthernet1/0/0]interface  GigabitEthernet 1/0/1
[FW1-GigabitEthernet1/0/1]
[FW1-GigabitEthernet1/0/1]service-manage enable
[FW1-GigabitEthernet1/0/1]
[FW1-GigabitEthernet1/0/1]service-manage ping permit
[FW1-GigabitEthernet1/0/1]
[FW1-GigabitEthernet1/0/1]quit
[FW1]
[FW1]firewall zone trust
[FW1-zone-trust]
[FW1-zone-trust]add  interface  GigabitEthernet 1/0/1
[FW1-zone-trust]
[FW1-zone-trust]firewall zone untrust
[FW1-zone-untrust]
[FW1-zone-untrust]add  interface  GigabitEthernet 1/0/0
[FW1-zone-untrust]
[FW1-zone-untrust]quit
[FW1]
```

二. 配置 DHCP,实现终端自动获取 IP 地址

1. 启用 DHCP 服务，配置对应地址池。

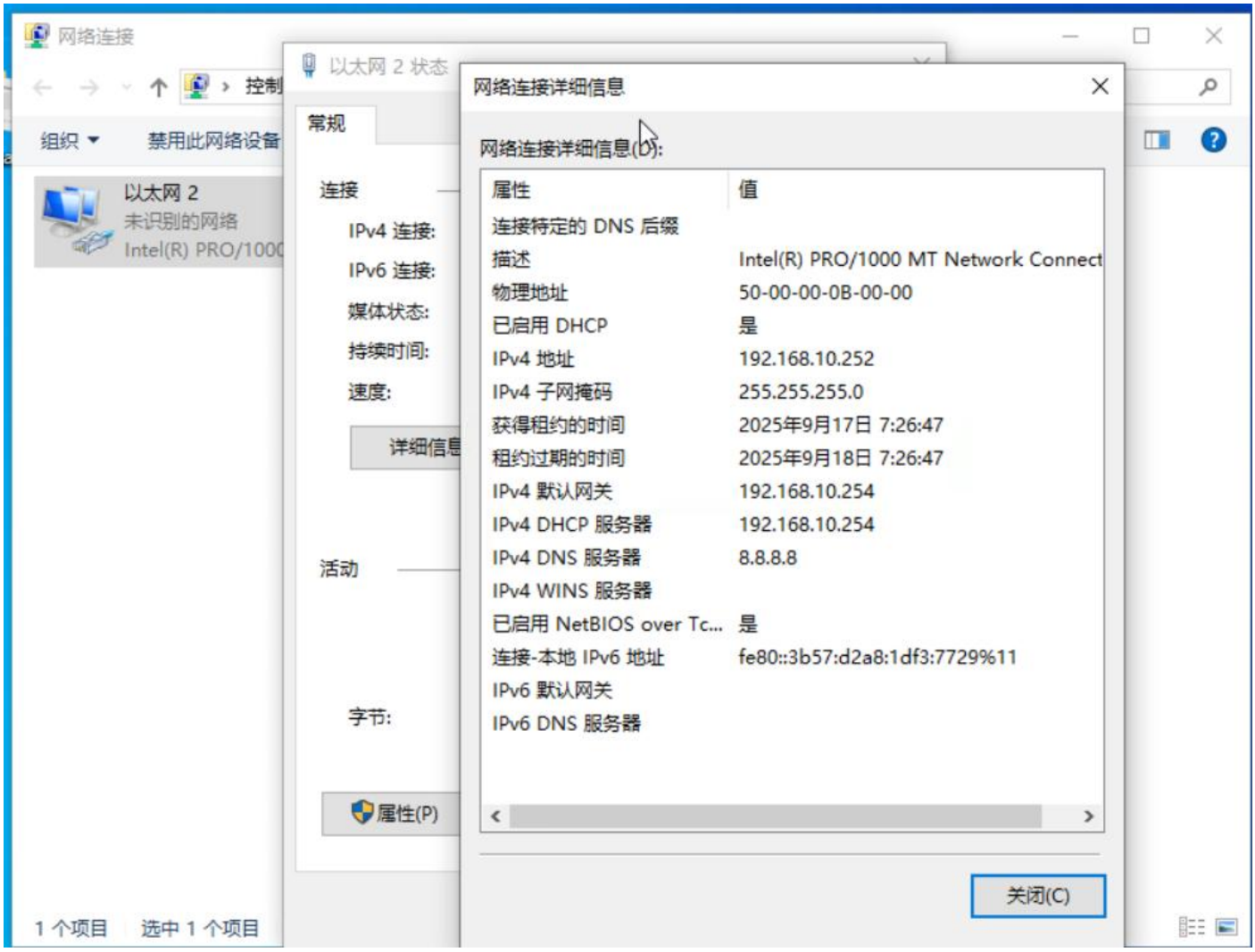
```
[SW1]
[SW1]dhcp enable
[SW1]
[SW1]ip pool vlan10
[SW1-ip-pool-vlan10]
[SW1-ip-pool-vlan10]gateway-list 192.168.10.254
[SW1-ip-pool-vlan10]
[SW1-ip-pool-vlan10]network 192.168.10.0 mask 255.255.255.0
[SW1-ip-pool-vlan10]
[SW1-ip-pool-vlan10]excluded-ip-address 192.168.10.1
[SW1-ip-pool-vlan10]
[SW1-ip-pool-vlan10]dns-list 8.8.8.8
[SW1-ip-pool-vlan10]
[SW1-ip-pool-vlan10]quit
[SW1]
[SW1]ip pool vlan20
[SW1-ip-pool-vlan20]
[SW1-ip-pool-vlan20]gateway-list 192.168.20.254
[SW1-ip-pool-vlan20]
[SW1-ip-pool-vlan20]network 192.168.20.0 mask 255.255.255.0
[SW1-ip-pool-vlan20]
[SW1-ip-pool-vlan20]dns-list 8.8.8.8
[SW1-ip-pool-vlan20]
[SW1-ip-pool-vlan20]quit
[SW1]
```

2. 开启 DHCP 监听（snooping）功能，并对特定接口进行 DHCP 监听。

```
[SW3]dhcp enable
[SW3]
[SW3]dhcp snooping enable
[SW3]
[SW3]int g1/0/2
[SW3-GE1/0/2]
[SW3-GE1/0/2]dhcp snooping enable
[SW3-GE1/0/2]
[SW3-GE1/0/2]int g1/0/4
[SW3-GE1/0/4]
[SW3-GE1/0/4]dhcp snooping enable
[SW3-GE1/0/4]
[SW3-GE1/0/4]quit
[SW3]
[SW3]int g1/0/0 # 连接合法 DHCP 服务器的接口
[SW3-GE1/0/0]
[SW3-GE1/0/0]dhcp snooping trusted # 设置为信任接口
[SW3-GE1/0/0]
```

```
[SW2]
[SW2]dhcp enable
[SW2]
[SW2]dhcp snooping  enable
[SW2]
[SW2]int g 1/0/2
[SW2-GE1/0/2]
[SW2-GE1/0/2]dhcp snooping  enable
[SW2-GE1/0/2]
[SW2-GE1/0/2]int g 1/0/0
[SW2-GE1/0/0]
[SW2-GE1/0/0]dhcp snooping  trusted
[SW2-GE1/0/0]
[SW2-GE1/0/0]quit
[SW2]
```

3. 验证，PC1 已成功自动获取 IP 地址。




三. 配置源 NAT，实现内网用户访问互联网。

1. 配置安全策略，允许从 trust 区域发往 untrust 区域的流量通过防火墙。

```
[FW1]
[FW1]security-policy
[FW1-policy-security]
[FW1-policy-security]rule name internet
[FW1-policy-security-rule-internet]
[FW1-policy-security-rule-internet]source-zone trust
[FW1-policy-security-rule-internet]
[FW1-policy-security-rule-internet]destination-zone untrust
[FW1-policy-security-rule-internet]
[FW1-policy-security-rule-internet]action  permit
[FW1-policy-security-rule-internet]
[FW1-policy-security-rule-internet]quit
[FW1-policy-security]
```

2. 配置 NAT 地址池（用于网络地址转换），实现内网地址向外网地址的映射。

```
[FW1]
[FW1]nat address-group internet
[FW1-address-group-internet]
[FW1-address-group-internet]mode  pat # 设置NAT 转换模式为PAT，端口地址转换
[FW1-address-group-internet]
[FW1-address-group-internet]section 0 200.1.1.1 200.1.1.10 # 地址组
[FW1-address-group-internet]
[FW1-address-group-internet]srcip-car-num 256 # 单个源 IP 最多可建立 256 连接
[FW1-address-group-internet]
[FW1-address-group-internet]route  enable # 启用地址组的路由功能
[FW1-address-group-internet]
[FW1-address-group-internet]quit
[FW1]
```

3. 配置 NAT 策略（网络地址转换策略），用于控制内网到外网的地址转换规则。

```
[FW1]nat-policy
[FW1-policy-nat]
[FW1-policy-nat]rule name internet
[FW1-policy-nat-rule-internet]
[FW1-policy-nat-rule-internet]source-zone trust
[FW1-policy-nat-rule-internet]
[FW1-policy-nat-rule-internet]destination-zone untrust
[FW1-policy-nat-rule-internet]
[FW1-policy-nat-rule-internet]action source-nat address-group internet
[FW1-policy-nat-rule-internet]# 指定对匹配该规则的流量采取 源 NAT 转换，并使用名为internet的 NAT 地址组
[FW1-policy-nat-rule-internet]quit
[FW1-policy-nat]
```

4. 由于地址池网段与接口 IP 不是在同一网段，配置静态路由发往地址池网段的流量，转发到下一跳。

```
[internet]
[internet]ip route-static 200.1.1.0 24 100.1.1.1
[internet]
```

5. 当地址池与出接口 IP 不在同一网段，将 nat 地址池地址配置为黑洞路由。

```
[FW1]
[FW1]ip route-static 200.1.1.0 24 NULL 0
[FW1]
```

四. 配置 VPN, 实现总部与分支内网互访。

1. 配置 GRE（通用路由封装）隧道以及静态路由，静态路由下一跳为 Tunnel0/0/0 接口。

```
[R1]
[R1]interface Tunnel0/0/0
[R1-Tunnel0/0/0]
[R1-Tunnel0/0/0]ip address 10.2.1.1 30
[R1-Tunnel0/0/0]
[R1-Tunnel0/0/0]tunnel-protocol gre
[R1-Tunnel0/0/0]
[R1-Tunnel0/0/0]source 100.1.1.6
[R1-Tunnel0/0/0]
[R1-Tunnel0/0/0]destination 100.1.1.1
[R1-Tunnel0/0/0]
[R1-Tunnel0/0/0]quit
[R1]
[R1]
[R1]ip route-static 192.168.10.0 24 tunnel0/0/0
[R1]
[R1]
```

2. 配置 IPSec（互联网协议安全）提案，定义 IPSec 通信的安全参数。

```
[R1]
[R1]ipse proposal tran1
[R1-ipsec-proposal-tran1]
[R1-ipsec-proposal-tran1]encapsulation-mode tunnel
[R1-ipsec-proposal-tran1]
[R1-ipsec-proposal-tran1]transform esp
[R1-ipsec-proposal-tran1]
[R1-ipsec-proposal-tran1]esp authentication-algorithm sha2-256
[R1-ipsec-proposal-tran1]
[R1-ipsec-proposal-tran1]esp encryption-algorithm aes-256
[R1-ipsec-proposal-tran1]
```

3. 配置 IKE 提案，用于 IPSec 等需要安全密钥交换的场景。

```
[R1]
[R1]ike proposal 5
[R1-ike-proposal-5]
[R1-ike-proposal-5]authentication-method pre-share
[R1-ike-proposal-5]
[R1-ike-proposal-5]encryption-algorithm aes-256
[R1-ike-proposal-5]
[R1-ike-proposal-5]authentication-algorithm sha2-256
[R1-ike-proposal-5]
[R1-ike-proposal-5]dh group14
[R1-ike-proposal-5]
```

4. 配置 IKE 对等体。

```
[R1]
[R1]ike peer spub
[R1-ike-peer-spub]
[R1-ike-peer-spub]undo version 2
Warning: This operation may delete the corresponding SA, Continue? [Y/N]:y
[R1-ike-peer-spub]
[R1-ike-peer-spub]exchange-mode main
[R1-ike-peer-spub]
[R1-ike-peer-spub]ike-proposal 5
[R1-ike-peer-spub]
[R1-ike-peer-spub]remote-address 100.1.1.1
[R1-ike-peer-spub]
[R1-ike-peer-spub]pre-shared-key cipher Key-1122
[R1-ike-peer-spub]
```

5. 配置访问控制列表（ACL）和 IPSec 策略，用于实现基于 IPSec 的安全通信

```
[R1]acl 3000
[R1-acl-adv-3000]
[R1-acl-adv-3000]rule permit ip source 100.1.1.6 0 destination 100.1.1.1 0
[R1-acl-adv-3000]
[R1-acl-adv-3000]quit
[R1]
[R1]ipsec policy map1 1 isakmp
Info: The ISAKMP policy sequence number should be smaller than the template policy sequence number in t
ISAKMP policy does not take effect.
[R1-ipsec-policy-isakmp-map1-1]
[R1-ipsec-policy-isakmp-map1-1]ike-peer spub
[R1-ipsec-policy-isakmp-map1-1]
[R1-ipsec-policy-isakmp-map1-1]proposal  tran1
[R1-ipsec-policy-isakmp-map1-1]
[R1-ipsec-policy-isakmp-map1-1]security acl 3000
[R1-ipsec-policy-isakmp-map1-1]
[R1-ipsec-policy-isakmp-map1-1]quit
```

6. 在路由器（R1）的 GigabitEthernet 0/0/0 接口上应用之前配置的 IPSec 策略。

```
[R1]
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]
[R1-GigabitEthernet0/0/0]ipsec policy map1
[R1-GigabitEthernet0/0/0]
```

7. 防火墙（FW1）配置隧道接口及安全区域。

```
[FW1]
[FW1]interface  Tunnel 0
[FW1-Tunnel0]
[FW1-Tunnel0]ip address  10.2.1.2 30
[FW1-Tunnel0]
[FW1-Tunnel0]quit
[FW1]
[FW1]firewall  zone  untrust
[FW1-zone-untrust]
[FW1-zone-untrust]add interface Tunnel 0
[FW1-zone-untrust]
```

8. 配置 untrust 和 trust 之间的安全策略，保证分支和总部能够进行互访

```
[FW1]security-policy
[FW1-policy-security]
[FW1-policy-security]rule name vpn1
[FW1-policy-security-rule-vpn1]
[FW1-policy-security-rule-vpn1]source-zone  untrust
[FW1-policy-security-rule-vpn1]
[FW1-policy-security-rule-vpn1]destination-zone trust
[FW1-policy-security-rule-vpn1]
[FW1-policy-security-rule-vpn1]source-address 192.168.30.0 24
[FW1-policy-security-rule-vpn1]
[FW1-policy-security-rule-vpn1]destination-address 192.168.10.0 24
[FW1-policy-security-rule-vpn1]
[FW1-policy-security-rule-vpn1]act permit
[FW1-policy-security-rule-vpn1]
```

9. 配置 local 与 untrust 之间的安全策略，保证防火墙能够发起和接收 IPSec 隧道建立请求，源、

目的 IP 地址为两端的出口公网地址。

```
[FW1-policy-security]
[FW1-policy-security]rule name vpn2
[FW1-policy-security-rule-vpn2]
[FW1-policy-security-rule-vpn2]source-zone  local
[FW1-policy-security-rule-vpn2]
[FW1-policy-security-rule-vpn2]destination-zone untrust
[FW1-policy-security-rule-vpn2]
[FW1-policy-security-rule-vpn2]source-address 100.1.1.1 32
[FW1-policy-security-rule-vpn2]
[FW1-policy-security-rule-vpn2]destination-address 100.1.1.6 32
[FW1-policy-security-rule-vpn2]
[FW1-policy-security-rule-vpn2]action  permit
[FW1-policy-security-rule-vpn2]
```

```
[FW1-policy-security]
[FW1-policy-security]rule name vpn3
[FW1-policy-security-rule-vpn3]
[FW1-policy-security-rule-vpn3]source-zone  untrust
[FW1-policy-security-rule-vpn3]
[FW1-policy-security-rule-vpn3]destination-zone local
[FW1-policy-security-rule-vpn3]
[FW1-policy-security-rule-vpn3]source-address 100.1.1.6 32
[FW1-policy-security-rule-vpn3]
[FW1-policy-security-rule-vpn3]destination-address 100.1.1.1 32
[FW1-policy-security-rule-vpn3]
[FW1-policy-security-rule-vpn3]action  permit
[FW1-policy-security-rule-vpn3]
[FW1-policy-security-rule-vpn3]quit
[FW1-policy-security]
```

10. 配置 GRE Tunnel 接口。

```
[FW1]
[FW1]interface  Tunnel 0
[FW1-Tunnel0]
[FW1-Tunnel0]tunnel-protocol gre
[FW1-Tunnel0]
[FW1-Tunnel0]source 100.1.1.1
[FW1-Tunnel0]
[FW1-Tunnel0]destination 100.1.1.6
[FW1-Tunnel0]
[FW1-Tunnel0]quit
[FW1]
[FW1]ip route-static 192.168.30.0 24 Tunnel 0
[FW1]
```

11. 防火墙配置 IPSec 安全提议。

```
[FW1]
[FW1]ipsec proposal tran1
[FW1-ipsec-proposal-tran1]
[FW1-ipsec-proposal-tran1]encapsulation-mode  tunnel
[FW1-ipsec-proposal-tran1]
[FW1-ipsec-proposal-tran1]transform  esp
[FW1-ipsec-proposal-tran1]
[FW1-ipsec-proposal-tran1]esp  authentication-algorithm  sha2-256
[FW1-ipsec-proposal-tran1]
[FW1-ipsec-proposal-tran1]esp encryption-algorithm  aes-256
[FW1-ipsec-proposal-tran1]
```

12. 配置 IKE 安全提议。

```
[FW1]
[FW1]ike proposal 5
[FW1-ike-proposal-5]
[FW1-ike-proposal-5]authentication-method pre-share
[FW1-ike-proposal-5]
[FW1-ike-proposal-5]encryption-algorithm  aes-256
[FW1-ike-proposal-5]
[FW1-ike-proposal-5]authentication-algorithm sha2-256
[FW1-ike-proposal-5]
[FW1-ike-proposal-5]dh  group14
[FW1-ike-proposal-5]
[FW1-ike-proposal-5]quit
[FW1]
```

13. 配置 IKE 对等体。

```
[FW1]
[FW1]ike peer  spub
[FW1-ike-peer-spub]
[FW1-ike-peer-spub]undo version  2
Warning: This operation may delete the corresponding SA, Continue? [Y/N]:y
[FW1-ike-peer-spub]
[FW1-ike-peer-spub]exchange-mode main
[FW1-ike-peer-spub]
[FW1-ike-peer-spub]ike-proposal 5
[FW1-ike-peer-spub]
[FW1-ike-peer-spub]pre-shared-key  Key-1122
[FW1-ike-peer-spub]
[FW1-ike-peer-spub]remote-address 100.1.1.6
[FW1-ike-peer-spub]
[FW1-ike-peer-spub]quit
[FW1]
```

14. 配置 IPSec 策略。

```
[FW1]
[FW1]acl 3000
[FW1-acl-adv-3000]
[FW1-acl-adv-3000]rule  permit  ip source 100.1.1.1 0 destination 100.1.1.6 0
[FW1-acl-adv-3000]
[FW1-acl-adv-3000]quit
[FW1]
[FW1]ipsec policy map1 1 isakmp
[FW1-ipsec-policy-isakmp-map1-1]
[FW1-ipsec-policy-isakmp-map1-1]ike-peer spub
[FW1-ipsec-policy-isakmp-map1-1]
[FW1-ipsec-policy-isakmp-map1-1]proposal tran1
[FW1-ipsec-policy-isakmp-map1-1]
[FW1-ipsec-policy-isakmp-map1-1]security acl 3000
[FW1-ipsec-policy-isakmp-map1-1]
[FW1-ipsec-policy-isakmp-map1-1]quit
```

15. 在接口上应用 IPSec 策略。

```
[FW1]
[FW1]interface  GigabitEthernet  1/0/0
[FW1-GigabitEthernet1/0/0]
[FW1-GigabitEthernet1/0/0]ipsec  policy  map1
[FW1-GigabitEthernet1/0/0]
```

16. 该版本路由器与防火墙通过 SHA2 认证算法进行 IPSec 对接时，会存在 SHA2 算法不一致的问题。SHA2 算法不一致会造成报文加解密失败，进而造成业务不通的现象。所以在路由器上配置和对端相同的 SHA2 加解密方式。

```
[R1]
[R1]ipsec authentication sha2 compatible enable
[R1]
```

17. 查看隧道建立。分支主机能够支持访问总部。

```
[R1]display ike sa
 Conn-ID    Peer                                        VPN             Flag(s)              Phase  RemoteType  RemoteID
-----------------------------------------------------------------------------------------------------------------------------
 142        100.1.1.1/500                                               RD|ST|A              v1:2   IP          100.1.1.1
 141        100.1.1.1/500                                               RD|ST|A              v1:1   IP          100.1.1.1

  Number of IKE SA : 2
-----------------------------------------------------------------------------------------------------------------------------

 Flag Description:
 RD--READY   ST--STAYALIVE   RL--REPLACED   FD--FADING   TO--TIMEOUT
 HRT--HEARTBEAT   LKG--LAST KNOWN GOOD SEQ NO.   BCK--BACKED UP
 M--ACTIVE   S--STANDBY   A--ALONE  NEG--NEGOTIATING
```

```
VPCS> show

NAME   IP/MASK              GATEWAY                           GATEWAY
VPCS1  192.168.30.1/24      192.168.30.254
       fe80::250:79ff:fe66:6809/64

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=61 time=4.259 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=61 time=3.541 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=61 time=3.255 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=61 time=3.878 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=61 time=3.384 ms
```

五. 配置 NAT server,实现内部服务器对外网的访问服务。

1. 配置安全策略，指定源安全区域为 untrust，目的安全区域为 dmz 。

```
[FW1]
[FW1]security-policy
[FW1-policy-security]
[FW1-policy-security]rule name natserver
[FW1-policy-security-rule-natserver]
[FW1-policy-security-rule-natserver]source-zone untrust
[FW1-policy-security-rule-natserver]
[FW1-policy-security-rule-natserver]destination-zone dmz
[FW1-policy-security-rule-natserver]
[FW1-policy-security-rule-natserver]destination-address 172.16.0.1 32
[FW1-policy-security-rule-natserver]
[FW1-policy-security-rule-natserver]action permit
[FW1-policy-security-rule-natserver]
[FW1-policy-security-rule-natserver]quit
[FW1-policy-security]
```

2. 将内部服务器地址映射到公网地址的的 8080 端口。

```
[FW1]
[FW1]nat server  yingshe protocol tcp  global  100.1.1.1 8080 inside  172.16.0.1 www
[FW1]
```

六. 配置 DDOS 攻击防范。

1. 配置攻击防范参数。

```
[FW1]
[FW1]interface  GigabitEthernet  1/0/0
[FW1-GigabitEthernet1/0/0]
[FW1-GigabitEthernet1/0/0]anti-ddos  flow-statistic enable
[FW1-GigabitEthernet1/0/0]  # 启用该接口的 Anti - DDoS 流量统计功能
[FW1-GigabitEthernet1/0/0]quit
[FW1]                  # 设置防火墙的 DDoS 防护模式为 （检测 - 清洗模式）
[FW1]ddos-mode detect-clean
[FW1]
```

2. 配置阈值学习功能。

```
[FW1]
[FW1]anti-ddos  baseline-learn  start
[FW1]  # 启动防火墙的 DDoS 基线学习功能，方便判断异常流量
[FW1]
[FW1]anti-ddos  baseline-learn  tolerance-value 100
[FW1]
[FW1]          # 设置 DDoS 基线学习的容忍值为 100
[FW1]anti-ddos  baseline-learn  apply
[FW1]
[FW1]          # 应用 DDoS 基线学习的配置
[FW1]
```

3. 开启攻击防范功能。

```
[FW1]
[FW1]anti-ddos  syn-flood  source-detect
[FW1]# 针对 SYN Flood(同步洪流)攻击，开启基于源的检测功能
[FW1]
[FW1]anti-ddos  udp-flood  dynamic-fingerprint-learn
[FW1]
[FW1]# 针对 用户数据报协议洪流 攻击，开启动态指纹学习功能
[FW1]anti-ddos  udp-frag-flood  dynamic-fingerprint-learn
[FW1]
[FW1]# 针对 UDP 分片洪流 攻击，开启动态指纹学习功能
[FW1]anti-ddos http-flood  defend alert-rate 2000
[FW1]
[FW1]# 针对超文本传输协议洪流攻击，配置防御的告警速率为 2000
[FW1]
[FW1]anti-ddos  http-flood  source-detect mode  basic
[FW1]        # 针对 HTTP Flood 攻击，开启基于源的检测功能，
[FW1]                 且检测模式为 “basic（基础）”
[FW1]
```

六. 总结

工程师小李完成多方面网络配置：设备基础配置含命名、IP 设置、VLAN 划分、路由部署与安全区域划分，保障架构通畅；配置 DHCP 服务与监听，实现终端自动获 IP，避免冲突；源 NAT 策略解决内网访问外网地址转换问题；部署 VPN（GRE 隧道 + IPSec 加密），建立总部与分支安全通道；配置 NAT Server 实现外网访问内部服务器；设 DDoS 防范策略抵御攻击。这些配置解决公司原网络

安全、访问、稳定性问题，保障网络安全、稳定、高效运行。